

February 11, 2026

Continuing with limits

e.g. Show that $\lim_{(x,y) \rightarrow (1,2)} (5x-3, x+2y) = (2,5)$.

Fix any arbitrary $\varepsilon > 0$.

We want to find $\delta > 0$ such that

$$\|(5x-3, x+2y) - (2,5)\| < \varepsilon$$

whenever $0 < \|(x,y) - (1,2)\| < \delta$.

Let's rewrite this and use the triangle inequality

$$\begin{aligned} \|(5x-3-2, x+2y-5)\| &= \|(5x-5, 0) + (0, x+2y-5)\| \\ &\leq \|(5x-5, 0)\| + \|(0, x+2y-5)\| = |5x-5| + |x+2y-5| \end{aligned}$$

iff $\|(x,y) - (1,2)\| < \delta$, then $|x-1| < \delta$ and $|y-2| < \delta$

$$\begin{aligned} \text{So } |5x-5| + |x+2y-5| &= 5|x-1| + |x-1 + 2(y-2)| \\ &\leq 5|x-1| + |x-1| + 2|y-2| < 5\delta + \delta + 2\delta \\ &= 8\delta \end{aligned}$$

We can pick $\delta = \varepsilon/8$.

Let's verify: if $\|(x,y) - (1,2)\| < \varepsilon/8$, then

$|x-1| < \varepsilon/8$ and $|y-2| < \varepsilon/8$, so

$$\begin{aligned} \|(5x-3, x+2y) - (2,5)\| &= \|(5x-5, x+2y-5)\| \leq |5x-5| + |x+2y-5| \\ &\leq 5|x-1| + |x-1| + 2|y-2| < \frac{5\varepsilon}{8} + \frac{\varepsilon}{8} + 2\frac{\varepsilon}{8} = \varepsilon \end{aligned}$$

morale: Limits for vector functions exist if and only if the limit exists for each component function

Theorem: Suppose $F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a function with components $f_1, \dots, f_n$
 (recall this means $F(x_1, \dots, x_m) = (f_1(x_1, \dots, x_m), \dots, f_n(x_1, \dots, x_m))$)
 then: $\lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) = \vec{L}$ (where $\vec{L} = (L_1, \dots, L_n)$) if and only if $\lim_{\vec{x} \rightarrow \vec{a}} f_i(\vec{x}) = L_i$ for all $i = 1, \dots, n$.

Limit laws

Suppose: • $F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $\lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) = \vec{L}$

• $G: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $\lim_{\vec{x} \rightarrow \vec{a}} G(\vec{x}) = \vec{M}$

• $f: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ and $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = c$

• $g: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ and $\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) = d$

then:

$$1) \lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) + G(\vec{x}) = \lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) + \lim_{\vec{x} \rightarrow \vec{a}} G(\vec{x}) = \vec{L} + \vec{M}$$

$$2) \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) F(\vec{x}) = \left(\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) \right) \left(\lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) \right) = c \vec{L}$$

$$3) \text{ if } h \in \mathbb{R}, \text{ then } \lim_{\vec{x} \rightarrow \vec{a}} h F(\vec{x}) = h \lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) = h \vec{L}$$

$$4) \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) g(\vec{x}) = cd$$

$$5) \lim_{\vec{x} \rightarrow \vec{a}} \frac{f(\vec{x})}{g(\vec{x})} = \frac{c}{d} \text{ as long as } d \neq 0 \text{ and } g(\vec{x}) \neq 0$$

e.g. $\lim_{(x,y) \rightarrow (1,3)} 3\left(4x^2 + y, \frac{x+y}{y}\right) + \left(\frac{y^2 x}{x-3}, x^4 y^3\right) =$
 $3(4(1^2) + 3, \frac{1+3}{3}) + \left(\frac{3^2 \cdot 1}{1-3}, 1^4 3^3\right) = 3(7, \frac{4}{3}) + (-4, 27) = (17, 37)$

recall in Calc I we could show $\lim_{x \rightarrow a} f(x)$ DNE
by showing $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$

we can do the "same" in Calc III only instead of
approaching from the left and approaching from the right
we can approach from arbitrary paths.

e.g. $f(x, y) = \frac{xy}{x^2 + y^2}$ show that $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ DNE :
plug in? $\frac{0 \cdot 0}{0^2 + 0^2}$ doesn't make sense ($\div 0$)

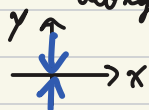
simplest paths (in general) are straight lines like:

1) $x=0 \quad y \rightarrow 0$

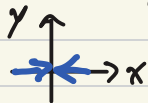
2) $y=0 \quad x \rightarrow 0$

3) $x=y \rightarrow 0$

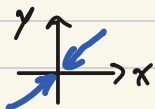
1) $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{y \rightarrow 0} f(0, y) = \lim_{y \rightarrow 0} \frac{0 \cdot y}{0 + y^2} = \lim_{y \rightarrow 0} \frac{0}{y^2} = 0$
along $x=0$



2) $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{x \rightarrow 0} f(x, 0) = \lim_{x \rightarrow 0} \frac{x \cdot 0}{x^2 + 0} = 0$
along $y=0$



3) $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{x \rightarrow 0} f(x, x) = \lim_{x \rightarrow 0} \frac{x^2}{x^2 + x^2} = \lim_{x \rightarrow 0} \frac{x^2}{2x^2} = \frac{1}{2}$
along $x=y$



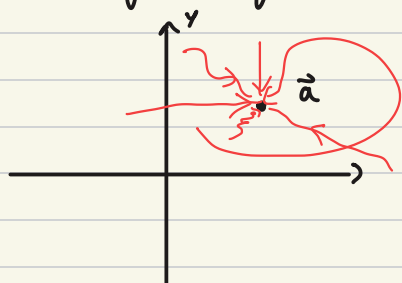
Summing up:

since $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) \neq \lim_{(x, y) \rightarrow (0, 0)} f(x, y)$
along $x=0$ along $x=y$

then
the limit DNE

Important !!

You can check along paths to show the limit DNE, but
checking along paths does not tell you if the limit does exist



(you would need to check infinitely many paths)

e.g. $\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x+y} \quad ?$

Let's take the path $x=0$

$$\begin{aligned} \lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } x=0}} f(x,y) &= \lim_{y \rightarrow 0} f(0,y) = \lim_{y \rightarrow 0} \frac{0-y}{0+y} \\ &= \lim_{y \rightarrow 0} -1 \cdot \frac{y}{y} = -1 \end{aligned}$$

Let's take the path $x=y$

$$\begin{aligned} \lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } x=y}} \frac{x-y}{x+y} &= \lim_{x \rightarrow 0} f(x,x) = \lim_{x \rightarrow 0} \frac{x-x}{x+x} = \lim_{x \rightarrow 0} \frac{0}{2x} \\ &= 0 \end{aligned}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x+y} \quad \text{DNE}$$

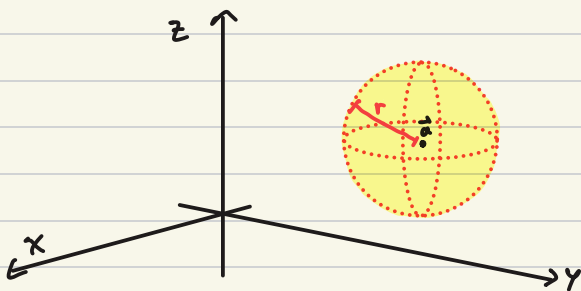
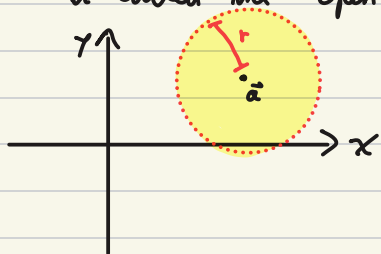
Continuity

recall from Calc I:

- open intervals : (a, b)
- closed intervals : $[a, b]$
- half open half closed : $[a, b)$ or $(a, b]$

$$(a, b) = (c-r, c+r) = \{x \in \mathbb{R} : |x-c| < r\} \quad \text{open interval}$$

def: For $\vec{a}$ in $\mathbb{R}^n$, the set $B_r(\vec{a}) := \{\vec{x} \in \mathbb{R}^n \mid \|\vec{x} - \vec{a}\| < r\}$ is called the open ball in $\mathbb{R}^n$ with radius r centered at $\vec{a}$.



some topology ideas

def: Let $S \subseteq \mathbb{R}^n$

- 1) A point $\vec{p} \in \mathbb{R}^n$ is called a boundary point of S if every open ball centered at $\vec{p}$ contains some point in S and some point not in S .
- 2) S is called closed if it contains all of its boundary points.
- 3) S is called open if it contains none of its boundary points.
- 4) A point $\vec{p}$ in S is called an isolated point of S if there is $\delta > 0$ such that no point in S satisfies $0 < \|\vec{x} - \vec{p}\| < \delta$ (i.e. no point in S other than $\vec{p}$ is in the ball $B_\delta(\vec{p})$).

def: A function $f: \underline{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be continuous at $\vec{a} \in \underline{X}$ if either:

1) $\vec{a}$ is an isolated point, or

2) $f(\vec{a}) = \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x})$

e.g. $\underline{X} = \{ (x, y) : x > 0, y > 0 \} \cup \{ (-1, 2) \}$
 $f(x, y) = \frac{1}{xy}$

domain $\underline{X}$:

